

There are a total of five problems. You have to solve all the problems.

Problem 1 (CO1): DFA and Regular Languages (15 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

$$L_1 = \{w : \text{the number of 1s in } w \text{ is a multiple of three}\}$$

$$L_2 = \{w : \text{every 0 in } w \text{ is followed by at least one 1}\}$$

$$L_3 = \{0^m 1^n : m, n \geq 1\}$$

$$L_4 = L_1 \cap L_2 \cap L_3$$

$$L_5 = \{0\}$$

$$L_6 = \{111\}$$

Now solve the following problems.

- Give the state diagram of a DFA that recognizes L_1 . (3 points)
- Give the state diagram of a DFA that recognizes L_2 . (3 points)
- Write the shortest string in L_3 . (1 point)
- Give the state diagram of a DFA that recognizes L_3 . (3 points)
- If you are to use the "cross product" construction to obtain a DFA that recognizes L_4 , how many states will it have? (1 point)
- Write all strings of length less than ten in L_4 . (1 point)
- Give the state diagram of an NFA that recognizes L_4 using no more than five states. (2 points)
- Is $L_5 \circ L_6^*$ a subset of L_3 ? Why or why not? (1 point)

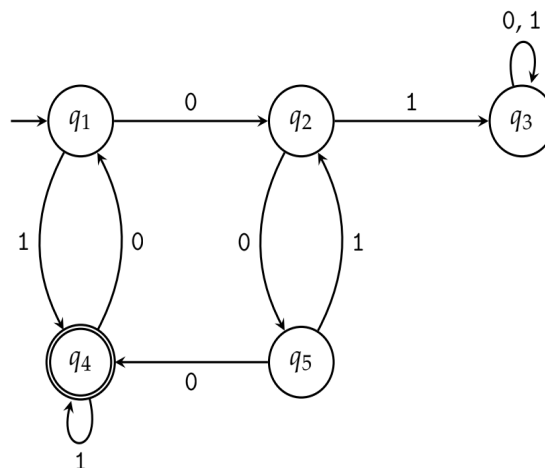
Problem 2 (CO2): Converting Regular Expressions to NFA (10 points)

Convert the following regular expression over $\Sigma = \{a, b, c\}$ into an equivalent NFA. Note that $R_1 + R_2$ is the same as $R_1 \cup R_2$.

$$c(b^*a + ac)^* + (c^* + bb^*)a$$

Problem 3 (CO2): Converting Finite Automata to Regular Expressions (10 points)

Convert the following finite automata into an equivalent regular expression using the state elimination method. You must eliminate the states in the following order: q_2, q_3, q_5, q_1, q_4 . Show each step of the process.



Problem 4 (CO1): Regular Expressions (10 points)

Let $\Sigma = \{a, b\}$. Consider the following languages over Σ .

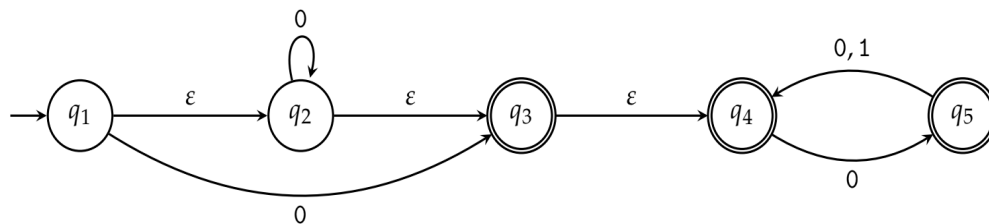
$$\begin{aligned} L_1 &= \{w : w \text{ contains no consecutive } a\} \\ L_2 &= \{w : w \text{ contains no consecutive } b\} \\ L_3 &= \{w : \text{last two symbols of } w \text{ are same}\} \\ L_4 &= \{w : \text{if } w \text{ starts with } b, \text{ it contains exactly two } a\} \\ L_5 &= L_1 \cap L_2 \end{aligned}$$

Now solve the following problems.

- Give a regular expression that generates $\overline{L_1}$. [Recall: $\overline{L}$ denotes the complement of the language L i.e., $\overline{L} = \Sigma^* - L$] (2 points)
- Give a regular expression that generates L_2 . (2 points)
- Give a regular expression that generates L_3 . (2 points)
- Give a regular expression that generates L_4 . (2 points)
- Give a regular expression that generates L_5 . (2 points)

Problem 5 (CO2): Subset Construction Method (5 points)

Consider the following NFA:



Now answer the following questions. [Note: You do not need to convert the given NFA into its equivalent DFA to answer the questions.]

- If you are to convert the given NFA into an equivalent DFA using the subset construction method, **What** is the maximum number of rejecting states that the equivalent DFA can have? (1 point)
- What** is the maximum number of accepting states that the equivalent DFA can have? (1 point)
- Write** the ϵ -closure of state q_1 in the given NFA. (1 point)
- What** is $\delta(\{q_4, q_5\}, 0)$ in the given NFA? [Recall: $\delta(\{q\}, a)$ outputs the set of states in which the NFA transitions when it is in state q and reads the symbol a .] (1 point)
- What** is $\delta(\{q_2\}, 1)$ in the given NFA? (1 point)

After you have finished the exam, please indicate where you stand on the smiley-face spectrum.

